Understanding the Fixed-Routing World:

the allocation bound, the residual cost, and the system's dynamics
(follow-up to the routing study; same scenario, no new regime)

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Summary

The routing study showed that fixing the order-routing rules removes 67–91% of the disruption cost and restores a 1.000 fill rate. Before changing anything about the scenario, we finished understanding it: (E1) *is there any value left in the distributor's allocation lever?* (E2) *what is the remaining 1.21M cost made of?* (E3) *how do the dynamics actually play out?* Findings: the allocation channel is worth **at most 10–12%**, the entire upside is captured by a *trivial static-priority rule* (at a fairness cost), and plausible-looking dynamic rules make things up to **56% worse**; the residual cost is mostly *bookkeeping at the dead factory* (41%) plus the unavoidable ∼10-period redirection transient, not a patient-service problem; and the one genuine physical inefficiency left — a post-recovery inventory glut that our own stale-order write-off *doubles* — is a small, fixable design artifact. Conclusion: **in this scenario there is no meaningful room left for learned policies**; what remains are two small deterministic refinements.

1 Setup

Table 1: All experiments run on the routing-fixed (rung c) baseline of the routing study: both HCs trust-split with trust[4] weights ($\delta' = 0.3$), stale-order write-off ($k = 5$), base-stock ordering everywhere. Reporting seeds 11–30, paired; phases pre 60–109 / during 110–157 / post 158–300; system-level cost (all six agents).

Experiment	**Design**
E1 allocation bound	DS allocation family {proportional, equal, priority-HC1, priority-HC2, backlog-priority, serve-captive}, applied at both DSs, closed loop. No tuned parameters. Max-over-family spread bounds the channel. Gate: the new code path under 'proportional' reproduces rung c *bit-for-bit* (60 columns × 300 periods).
E2 residual decomposition	Rung-c's 1.21M during+post cost, decomposed by agent × phase × (holding | backlog); glut quantified. Pure analysis of existing logs; totals cross-check against the routing study (pass).
E3 dynamics anatomy	Period-by-period trust, order routing, receipts, on-order ledger vs. counted, up-to targets, dead-factory queue. Pure analysis of existing logs.

2 E1 — The allocation bound

Table 2: Allocation family on the rung-c baseline. dp-cost = during+post system cost (urgent0; SEs $\leq$ 0.4% of means). Lost patients from the urgent20 config (episode total). Total units shipped are identical across rules — every difference is a closed-loop trust-feedback effect.

Allocation rule	dp-cost	vs. proportional	Lost patients
priority-HC1 / -HC2 (static)	**1,060k**	**−12.4%**	**445**
proportional (= rung c)	1,210k	—	510
equal	1,209k	−0.0%	509
serve-captive	1,517k	+25%	603
backlog-priority	1,881k	+56%	795

Three readings. **(i) The channel is second-order**: 10–12% at best, against routing's 67–91%. **(ii) The upside is trivial to capture and symmetric**: strict static priority wins, and it does not matter which HC is prioritized — what wins is *consistency* (serve one customer fully, keep its delivery/trust loop clean) rather than any clever targeting. The gain costs fairness: the non-prioritized HC's during fill drops to 0.957 vs. 0.981 (urgent20); proportional and equal are fairness-neutral. If fairness is a hard constraint, the channel's usable upside is $\approx$ zero. **(iii) The downside is** 5× **the upside**: backlog-priority — the "obvious" smart rule — alternates failures across both HCs, keeps both trust loops degraded, and lands 56% *worse* with more lost patients. For the eventual DRL comparison this bounds Mechanism 3 (allocation intelligence) tightly and asymmetrically: a learner exploring this space has little to win and much to lose.

3 E2 — What the remaining 1.21M is made of

Table 3: Rung-c during+post system cost by agent (urgent0, mean over seeds).

Agent	Cost	Dominant component
$\mathrm{MN}_{\mathrm{disrupted}}$	498k (41%)	orders queued at the shut factory (backlog cost, up to 48 periods)
$\mathrm{DS}_{\mathrm{disrupted}}$	345k (29%)	early orders + the ∼10-period redirection transient
$\mathrm{DS}_{\mathrm{healthy}}$	122k (10%)	the absorption transient
HCs (both)	183k (15%)	brief service dip (during) + recovery holding (post)
$\mathrm{MN}_{\mathrm{healthy}}$	61k (5%)	carrying the surge

None of this is a during-disruption service problem (fill is 1.000). Two observations for discussion. **The glut is real and partly self-inflicted**: post-disruption system inventory peaks at 3,930 vs. a pre-disruption mean of 332 — *double* the baseline's glut (2,051) — because the write-off lets $\mathrm{HC}_{\mathrm{equal}}$ re-order from the healthy chain while the written-off stale orders still deliver late. The cost (post holding 135k vs. 107k baseline) is ∼20× smaller than the backlog cost the write-off saves, but a recovery-aware refinement could keep the savings without the glut. **A modeling question for the advisor**: 41% of the residual is backlog cost accruing on undeliverable orders queued at a shut factory — an accounting convention, not a physical loss. Under a patient-facing lens the true residual is roughly half the headline number.

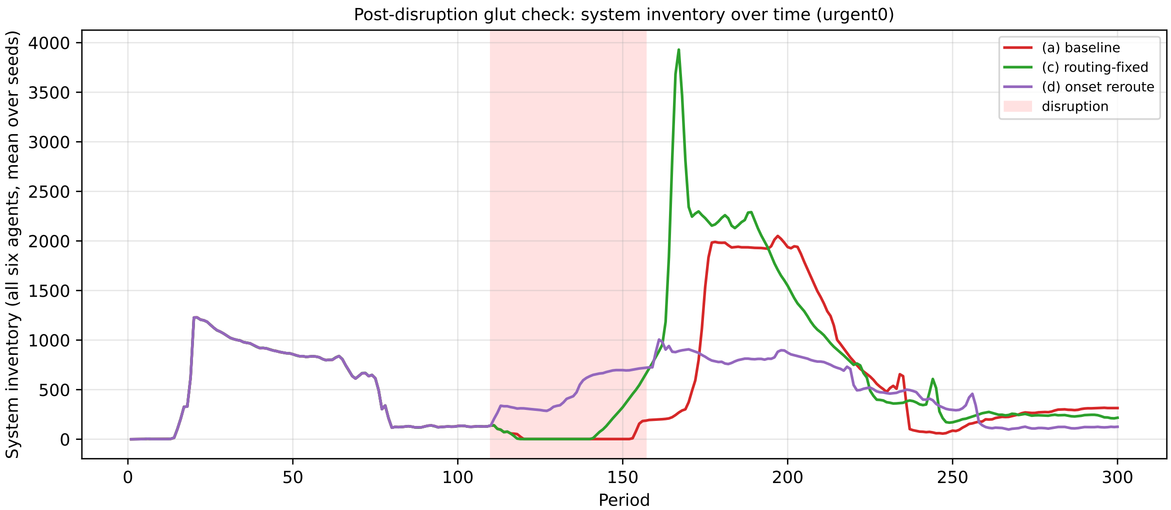

Figure 1: System inventory over time. The rung-c glut (green, peak 3,930) is double the baseline's (red) and almost six times the oracle-reroute rung's (purple) — the price of the accounting-only write-off, visible and fixable.

4 E3 — How the fixed-routing world actually moves

Figure 2: HC layer. (1) Trust collapses to ∼0.55 in ∼10 periods (the EMA floor — the trickle keeps delivering, so trust never reaches 0) and recovers ∼20 periods after the disruption. (2) Orders follow trust; the baseline (dotted) never moves. (3) Receipts follow orders with pipeline lag; the recovery-time spike of late deliveries from the dead chain is the glut arriving. (4) The write-off: the raw on-order ledger climbs to ∼800 while the ordering formula counts only ∼450 — the ignored gap is what frees $\mathrm{HC}_{\mathrm{equal}}$ to re-order from the healthy chain.

Figure 3: DS layer. (1) $\mathrm{DS}_{\mathrm{disrupted}}$'s backlog peaks at $\sim$1,000 (vs. 4,100 baseline); $\mathrm{DS}_{\mathrm{healthy}}$ absorbs with a brief $\sim$570 transient. (2) A counterintuitive finding: the *healthy* chain's up-to target balloons to $\sim$900 (its lead time stretches under the surge) while the dead chain's target *falls* with its rerouted demand. (3) $\mathrm{DS}_{\mathrm{disrupted}}$'s during-disruption orders are modest, but they queue at the shut factory for the whole window (the 441k); the post-recovery catch-up spike feeds the glut.

5 What we learned, and next steps

What we learned.

1. **The scenario is essentially solved by routing.** With routing fixed, fill rate is 1.000; no remaining cost is a patient-service problem.

2. **The allocation channel — the last DS-level lever, and the last place a DRL contribution could hide in this scenario — is bounded at 10–12%**, fully captured by a trivial static rule, bought at a fairness cost, with a 5× asymmetric downside for getting it wrong. There is no meaningful room left for learned policies here.

3. **The residual cost is mostly accounting** (41% dead-factory queue bookkeeping) plus an unavoidable ∼10-period transient; the one physical inefficiency (the glut) is a fixable artifact of our own write-off design.

4. **Two dynamics surprises** worth carrying: the healthy chain's up-to target is the one that balloons (not the dead chain's), and consistency beats cleverness in allocation because of the trust feedback loop.

Next steps (in order).

1. **Recovery-aware write-off refinement** (deterministic, small): suppress re-ordering as written-off pipeline approaches delivery — removes the glut, closes the main self-inflicted cost.

2. **Recompute the buffer sweep and clairvoyant ceiling on the rung-c baseline** — the gating step before any value-decomposition number goes in a paper.

3. **Write up the simple scenario completely** (routing study + this study). Only then consider robustness regimes (severity/duration sweeps), introduced because the write-up demands generality — not to give tools a job.

4. **When the corrected DRL arrives**: evaluate it against the new compound simple baseline — rung-c routing + static-priority allocation. The bar: beat 1.06M during+post cost (urgent0) / 445 lost patients (urgent20) without losing fairness.